

SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

Unit-wise Question Bank

Unit-1: Introduction to Networks

Course Outcome Mapped: CO-1 | Total Questions: 50

1	Define a computer network and state its key objectives.	CO-1
2	Explain the terms node, link, and protocol in the context of computer networks.	CO-1
3	What is a LAN? Give one real-life example.	CO-1
4	What is a MAN? Give one real-life example.	CO-1
5	What is a WAN? Give one real-life example.	CO-1
6	What is a PAN? Give one real-life example.	CO-1
7	Differentiate between LAN and WAN based on coverage area and speed.	CO-1
8	Compare MAN and WAN with suitable examples.	CO-1
9	List two advantages and two disadvantages of a Bus topology.	CO-1
10	Draw and explain the Bus network topology.	CO-1
11	Draw and explain the Star network topology.	CO-1
12	List two advantages and two disadvantages of a Star topology.	CO-1
13	Draw and explain the Ring network topology.	CO-1
14	Why is fault isolation difficult in a Ring topology?	CO-1
15	Draw and explain the Mesh network topology.	CO-1
16	Calculate the number of links required for a full Mesh topology with 8 nodes.	CO-1
17	What is a Hybrid topology? Give one real-world example of its use.	CO-1
18	Compare Star and Mesh topologies in terms of cost and reliability.	CO-1
19	Define Circuit Switching with a real-life example.	CO-1
20	List the three phases of a circuit-switched connection.	CO-1
21	Define Packet Switching with a real-life example.	CO-1
22	Differentiate between Circuit Switching and Packet Switching.	CO-1
23	Why is packet switching more suitable for the Internet than circuit switching?	CO-1
24	What is the OSI reference model? State the number of layers and their names in order.	CO-1
25	Explain the function of the Physical layer with one example device.	CO-1
26	Explain the function of the Data Link layer with one example protocol.	CO-1
27	Explain the function of the Network layer with one example protocol.	CO-1

28	Explain the function of the Transport layer with one example protocol.	CO-1
29	Explain the function of the Session layer with one example.	CO-1
30	Explain the function of the Presentation layer with one example.	CO-1
31	Explain the function of the Application layer with one example protocol.	CO-1
32	List the four layers of the TCP/IP model.	CO-1
33	Compare the OSI model and TCP/IP model, listing two similarities and two differences.	CO-1
34	Identify the OSI layer at which a switch primarily operates. Justify your answer.	CO-1
35	Identify the OSI layer at which a router primarily operates. Justify your answer.	CO-1
36	Define latency and list its components.	CO-1
37	Define bandwidth and state its unit of measurement.	CO-1
38	Differentiate between bandwidth and throughput.	CO-1
39	Explain propagation delay and transmission delay with formulas.	CO-1
40	What is the bandwidth-delay product? Why is it important?	CO-1
41	Describe the construction and use of Twisted Pair cable.	CO-1
42	Differentiate between UTP and STP cables.	CO-1
43	Describe the construction and use of Coaxial cable.	CO-1
44	Describe the construction and use of Fiber Optic cable.	CO-1
45	Compare Twisted Pair, Coaxial, and Fiber Optic cables on cost and bandwidth.	CO-1
46	Explain the characteristics and applications of Radio waves.	CO-1
47	Explain the characteristics and applications of Microwaves.	CO-1
48	Explain the characteristics and applications of Infrared.	CO-1
49	Compare Guided and Unguided transmission media.	CO-1
50	A company wants to connect 5 branch offices with maximum fault tolerance. Suggest a suitable topology and justify with link calculation.	CO-1

Unit-2: Addressing

Course Outcome Mapped: CO-2 | Total Questions: 50

1	Define IPv4 addressing and state its address length.	CO-2
2	What is meant by address space in IPv4?	CO-2
3	Explain Classful IP addressing with class ranges of A, B and C.	CO-2
4	State the address range and default subnet mask of Class A.	CO-2
5	State the address range and default subnet mask of Class B.	CO-2
6	State the address range and default subnet mask of Class C.	CO-2
7	What is the purpose of Class D addresses?	CO-2
8	What is the purpose of Class E addresses?	CO-2
9	Define subnet mask and explain its role in an IP address.	CO-2
10	Given IP 172.16.5.10/16, identify the network ID and host ID.	CO-2
11	What is Classless Addressing? Why was it introduced?	CO-2
12	Explain CIDR notation with one example.	CO-2
13	What is VLSM? State one advantage over FLSM.	CO-2
14	Given network 192.168.5.0/24, divide it into subnets of 60 and 30 hosts using VLSM.	CO-2
15	Define supernetting with one numerical example.	CO-2
16	Combine 192.168.4.0/24 and 192.168.5.0/24 into a single supernet. Show the calculation.	CO-2
17	What is NAT? State its purpose.	CO-2
18	Differentiate between Static NAT and Dynamic NAT.	CO-2
19	What is PAT (Port Address Translation)? Give one real-life use case.	CO-2
20	List the commands used to configure static NAT on a router.	CO-2
21	Explain Static Routing with an example configuration command.	CO-2
22	Explain Default Routing with an example configuration command.	CO-2
23	Differentiate between Static Routing and Default Routing.	CO-2
24	When would an administrator prefer Default Routing over Static Routing?	CO-2
25	Calculate the number of usable hosts in a /27 subnet.	CO-2
26	Calculate the number of usable hosts in a /28 subnet.	CO-2
27	Find the broadcast address for the network 10.10.10.0/26.	CO-2
28	Find the first and last usable host address for 192.168.1.64/27.	CO-2
29	Convert the subnet mask 255.255.255.192 to CIDR notation.	CO-2
30	Convert /20 CIDR notation to its equivalent dotted decimal subnet mask.	CO-2

31	Explain how a router determines whether a destination is on the local network or a remote network.	CO-2
32	What is a private IP address range? Give one example range.	CO-2
33	Why is NAT necessary for private IP addresses to access the Internet?	CO-2
34	Explain the concept of address exhaustion in IPv4 and how VLSM helps mitigate it.	CO-2
35	Design a VLSM addressing scheme for a network with four subnets needing 100, 50, 25 and 10 hosts respectively from 172.16.0.0/22.	CO-2
36	Differentiate between Classful and Classless addressing in terms of routing table size.	CO-2
37	What is meant by 'longest prefix match' in the context of classless addressing?	CO-2
38	Explain how subnetting improves network security and performance.	CO-2
39	State two limitations of NAT.	CO-2
40	Explain the configuration steps to enable PAT on a Cisco router.	CO-2
41	Given a Class C network 200.10.10.0/24, subnet it to create 4 equal subnets. State the new subnet mask.	CO-2
42	Explain the difference between a network address and a broadcast address with an example.	CO-2
43	What is the loopback address in IPv4? State its typical use.	CO-2
44	Explain the significance of the subnet mask 255.255.255.255 (/32).	CO-2
45	Compare VLSM and supernetting in terms of purpose and outcome.	CO-2
46	Explain how Default Routing simplifies configuration in a stub network.	CO-2
47	What information does the 'show ip nat translations' command display?	CO-2
48	Explain the three components required to configure Dynamic NAT (ACL, pool, inside/outside interfaces).	CO-2
49	Why can two Class C networks not always be summarized into a single supernet? Explain with a counter-example.	CO-2
50	A network administrator has been allocated 10.0.0.0/22. Design subnets for 3 departments needing 200, 100 and 50 hosts using VLSM.	CO-2

Unit-3: Routing

Course Outcome Mapped: CO-3 | Total Questions: 50

1	Explain the process of forwarding an IP packet at a router.	CO-3
2	What is a routing table? List its key fields.	CO-3
3	Differentiate between Static Routing and Dynamic Routing.	CO-3
4	What is Default Routing? When is it used?	CO-3
5	Define Unicast routing and give one example protocol.	CO-3
6	Explain the working principle of Distance Vector Routing.	CO-3
7	What algorithm does Distance Vector Routing use? Briefly explain it.	CO-3
8	What is the 'count-to-infinity' problem in Distance Vector Routing?	CO-3
9	Explain split horizon and how it helps prevent routing loops.	CO-3
10	Explain the working principle of Link State Routing.	CO-3
11	What algorithm does Link State Routing use to compute shortest paths?	CO-3
12	Differentiate between Distance Vector and Link State routing algorithms.	CO-3
13	Explain Path Vector Routing and its use in inter-domain routing.	CO-3
14	How does Path Vector Routing prevent routing loops between autonomous systems?	CO-3
15	State the address length used in IPv6.	CO-3
16	Write the general format of an IPv6 address with one example.	CO-3
17	Explain the use of '::' notation in IPv6 addressing.	CO-3
18	List the different types of IPv6 addresses.	CO-3
19	What is an Anycast address in IPv6?	CO-3
20	Define multicasting and differentiate it from unicasting and broadcasting.	CO-3
21	What is the role of IGMP in IP multicasting?	CO-3
22	State the Class D address range used for IPv4 multicasting.	CO-3
23	What is RIP? State its maximum hop count limit.	CO-3
24	Differentiate between RIP V1 and RIP V2.	CO-3
25	Why does RIP V2 support VLSM while RIP V1 does not?	CO-3
26	What is the significance of the hop-count metric in RIP?	CO-3
27	Explain the concept of areas in OSPF.	CO-3
28	Why is Area 0 (backbone area) mandatory in OSPF?	CO-3
29	What is the metric used by OSPF to calculate the shortest path?	CO-3
30	Explain the role of the Hello protocol in OSPF neighbor discovery.	CO-3
31	What is a Link State Advertisement (LSA) in OSPF?	CO-3

32	Differentiate between Single Area and Multi Area OSPF.	CO-3
33	What is BGP? State whether it is an interior or exterior gateway protocol.	CO-3
34	Explain the AS-PATH attribute in BGP.	CO-3
35	Differentiate between eBGP and iBGP.	CO-3
36	What is EIGRP? State whether it is Cisco-proprietary or an open standard.	CO-3
37	What composite metric factors does EIGRP use to calculate path cost?	CO-3
38	Explain the DUAL algorithm used by EIGRP.	CO-3
39	Compare RIP, OSPF, BGP and EIGRP in terms of routing type (distance vector/link state/path vector).	CO-3
40	Why is OSPF preferred over RIP for large enterprise networks?	CO-3
41	State one advantage and one disadvantage of Static Routing compared to Dynamic Routing.	CO-3
42	Explain how a router chooses between multiple routes to the same destination (administrative distance).	CO-3
43	What is meant by convergence time in a routing protocol? Why is fast convergence important?	CO-3
44	Differentiate between Interior Gateway Protocols (IGP) and Exterior Gateway Protocols (EGP), giving one example of each.	CO-3
45	Explain how the routing table entry for a directly connected network differs from a learned route.	CO-3
46	What is route summarization, and how does it help reduce routing table size?	CO-3
47	Explain the significance of administrative distance when multiple routing protocols advertise the same route.	CO-3
48	Draw and explain a simple network with three routers running RIP, showing how routes converge.	CO-3
49	Explain how OSPF handles a link failure and recalculates routes.	CO-3
50	Compare the scalability of Distance Vector, Link State, and Path Vector routing protocols.	CO-3

Unit-4: Medium Access Control

Course Outcome Mapped: CO-4 | Total Questions: 50

1	What is Medium Access Control (MAC)? Why is it required?	CO-4
2	Explain Pure ALOHA and derive its maximum efficiency.	CO-4
3	Explain Slotted ALOHA and state why it is more efficient than Pure ALOHA.	CO-4
4	Compare Pure ALOHA and Slotted ALOHA.	CO-4
5	Explain CSMA (Carrier Sense Multiple Access) and its persistence strategies (1-persistent, non-persistent, p-persistent).	CO-4
6	Explain CSMA/CD with a flowchart.	CO-4
7	What is a collision in a shared medium network? How does CSMA/CD detect it?	CO-4
8	Explain the binary exponential backoff algorithm used in CSMA/CD.	CO-4
9	Explain CSMA/CA and how it differs from CSMA/CD.	CO-4
10	What is the hidden terminal problem? How does CSMA/CA address it?	CO-4
11	Explain the role of RTS/CTS frames in CSMA/CA.	CO-4
12	Describe the frame format and access method of Ethernet (IEEE 802.3).	CO-4
13	Explain the working principle of Token Ring (IEEE 802.5).	CO-4
14	Compare Ethernet and Token Ring in terms of access method and performance.	CO-4
15	Why has Token Ring largely been replaced by switched Ethernet?	CO-4
16	Explain Stop-and-Wait flow control with a timing diagram.	CO-4
17	State one limitation of Stop-and-Wait flow control.	CO-4
18	Explain Sliding Window flow control and define window size.	CO-4
19	Compare Stop-and-Wait and Sliding Window flow control in terms of efficiency.	CO-4
20	Explain Stop-and-Wait ARQ, including the role of the timer.	CO-4
21	What happens in Stop-and-Wait ARQ when an ACK is lost?	CO-4
22	Explain Go-Back-N ARQ with a suitable example.	CO-4
23	Explain Selective Repeat ARQ and how it differs from Go-Back-N.	CO-4
24	Compare Go-Back-N ARQ and Selective Repeat ARQ.	CO-4
25	Define error detection and error correction. State their difference.	CO-4
26	Explain the Parity Check method for error detection with an example.	CO-4
27	State one limitation of the simple parity check method.	CO-4
28	Explain two-dimensional (block) parity check and how it improves error detection.	CO-4
29	Explain the Checksum method for error detection.	CO-4
30	Compute the checksum for two 8-bit data words: 10101001 and 00111001.	CO-4

31	Explain the Cyclic Redundancy Check (CRC) method with the concept of a generator polynomial.	CO-4
32	Perform CRC division for data 1101011011 with generator polynomial 1011, and find the remainder.	CO-4
33	Explain Hamming code and state how many redundant bits are needed for 4 data bits.	CO-4
34	Encode the data word 1101 using Hamming code and show the final codeword.	CO-4
35	Explain how the Hamming code receiver detects and corrects a single-bit error.	CO-4
36	Differentiate between error detection codes and error correction codes.	CO-4
37	What is HDLC? List its three transfer modes.	CO-4
38	Explain the frame format of HDLC.	CO-4
39	What is PPP? List its two main sub-protocols (LCP and NCP).	CO-4
40	Differentiate between HDLC and PPP.	CO-4
41	Explain the purpose of a jam signal in CSMA/CD.	CO-4
42	What is the vulnerable period in Pure ALOHA and Slotted ALOHA?	CO-4
43	Explain why wireless LANs (IEEE 802.11) use CSMA/CA instead of CSMA/CD.	CO-4
44	Explain the difference between flow control and error control.	CO-4
45	What is meant by a 'window size' of 1 in Sliding Window protocol? What does it become equivalent to?	CO-4
46	Explain bit stuffing and its use in HDLC framing.	CO-4
47	State the authentication protocols supported by PPP (PAP and CHAP) and differentiate them.	CO-4
48	Explain how CRC provides better burst-error detection compared to checksum.	CO-4
49	A 7-bit Hamming code is received as 1010101. Check whether it contains an error and, if so, correct it.	CO-4
50	Compare the overhead and reliability trade-offs among Parity Check, Checksum, and CRC.	CO-4

Unit-5: Transport and Application Layer Protocols

Course Outcome Mapped: CO-5 | Total Questions: 50

1	What is a port number? State its size in bits and range.	CO-5
2	List five well-known port numbers along with their associated services.	CO-5
3	Differentiate between well-known, registered, and dynamic/private ports.	CO-5
4	Define a socket and explain how it uniquely identifies a connection.	CO-5
5	Explain the User Datagram Protocol (UDP) and its key characteristics.	CO-5
6	Describe the UDP header format and its fields.	CO-5
7	List two applications where UDP is preferred over TCP, with justification.	CO-5
8	Explain the Transmission Control Protocol (TCP) and its key characteristics.	CO-5
9	Describe the TCP header format and explain any four fields.	CO-5
10	Compare TCP and UDP in terms of reliability, ordering, and overhead.	CO-5
11	Explain the TCP three-way handshake with a diagram.	CO-5
12	Explain the TCP connection termination process (four-way handshake).	CO-5
13	What is meant by TCP flow control using the sliding window mechanism?	CO-5
14	Explain TCP congestion control and the role of slow start.	CO-5
15	What is the purpose of the TCP checksum field?	CO-5
16	Explain the significance of sequence and acknowledgement numbers in TCP.	CO-5
17	What is the World Wide Web (WWW)? Explain its basic architecture.	CO-5
18	Explain HTTP and its request-response model.	CO-5
19	List and explain any four HTTP methods (GET, POST, PUT, DELETE).	CO-5
20	Differentiate between HTTP and HTTPS.	CO-5
21	Explain the concept of HTTP status codes with two examples.	CO-5
22	What is a URL? Explain its components.	CO-5
23	Explain the File Transfer Protocol (FTP) and the ports it uses.	CO-5
24	Differentiate between FTP Active mode and Passive mode.	CO-5
25	Explain the two types of connections used in FTP.	CO-5
26	Describe the email architecture, including MUA and MTA.	CO-5
27	Explain the working of SMTP and the port it uses.	CO-5
28	Explain POP3 and its limitations for multi-device access.	CO-5
29	Explain IMAP and how it differs from POP3.	CO-5
30	Compare SMTP, POP3, and IMAP in terms of their roles.	CO-5
31	What is Telnet? Explain its working and the port it uses.	CO-5

32	State two security limitations of Telnet.	CO-5
33	Why has SSH largely replaced Telnet for remote administration?	CO-5
34	Explain the Domain Name System (DNS) and its purpose.	CO-5
35	Explain the hierarchical structure of the DNS namespace.	CO-5
36	What is a DNS resolver? Explain its role in name resolution.	CO-5
37	Differentiate between a recursive and an iterative DNS query.	CO-5
38	Explain the steps involved in resolving a domain name to an IP address.	CO-5
39	What is a DNS record? Explain the purpose of A, AAAA, MX, and CNAME records.	CO-5
40	Explain the role of a root server, TLD server, and authoritative server in DNS resolution.	CO-5
41	What is DNS caching, and how does it improve resolution performance?	CO-5
42	Compare connection-oriented and connectionless transport services with examples.	CO-5
43	Explain multiplexing and demultiplexing at the transport layer.	CO-5
44	What is the purpose of the TCP MSS (Maximum Segment Size)?	CO-5
45	Explain how TCP achieves reliable delivery despite using an unreliable network layer.	CO-5
46	Differentiate between persistent and non-persistent HTTP connections.	CO-5
47	Explain the purpose of cookies in HTTP-based web applications.	CO-5
48	State the difference between the World Wide Web and the Internet.	CO-5
49	Explain a real-world scenario where UDP would fail to be reliable and how the application layer compensates.	CO-5
50	Trace the complete path of data from a web browser request to a DNS server response, identifying each protocol involved (DNS, TCP, HTTP).	CO-5